Department of Aerospace Engineering
Faculty of Engineering and Architectural Science

Project 2: Flight Simulator Physics Engine

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Objective

This project provides an excellent opportunity for students to work in teams and design, implement and test a specialized engineering software system. The learning outcomes of this project include:

- Learning how to use MATLAB and Simulink.
- Connecting abstract mathematical and physical concepts learned in Calculus and Dynamics courses to a tangible physical system; that is aircraft, and its flight dynamics.
- Computer programming for an engineering application and building foundation for the future computer programming tasks.
- Developing an understanding of modular system design and learning how to interface a software module with other software modules to create a standalone system.

Client Statement

An aircraft manufacturing company has designed a new airliner. The aircraft is a narrow-body twinjet airliner with a capacity of 240-280 passengers competing with Boeing 737 and Airbus A320 series.

The company has formed a "Flight Simulation Division (FSD)" to develop a flight simulator for the aircraft that will be used for pilot training. The FSD consist of three groups:

- *Physics Engine:* comprised by aerospace engineers who are responsible for developing the "brain" of the simulator; that is, how aircraft dynamic behaviour in flight can be realistically simulated in a virtual environment.
- *Graphics:* comprised by software engineers who are responsible for developing a graphical environment that will represent the "world" in the simulator.
- *User Interface:* comprised by electrical and mechanical engineers who are responsible for integrating joysticks, processors, displays, and the simulator software.

As an aerospace engineer, you are hired to work in the Physics Engine group. The primary task is to implement equations that will simulate six degrees-of-freedom free flight for the target aircraft with the following requirements:

- The equations must consider the throttle settings and deflections on primary control surfaces (aileron, elevator, and rudder).
- The equations must predict the states of all three Euler angles (roll, pitch, yaw), and body rates.
- The equations must predict the longitudinal, lateral, and vertical speed of aircraft.
- The equations must predict the geodetic coordinates of airplane (latitude, longitude, altitude).
- To ensure an acceptable level of fidelity for the simulations, flat-Earth equations or more accurate. models must be implemented, and the numerical integration method must be the fourth order Runge-Kutta or more accurate methods.

The aircraft design team has previously modeled the aircraft aerodynamic characteristics using wind-tunnel simulations, computational fluid dynamics (CFD), and flight tests. The information about the aircraft aerodynamic coefficients is provided in the supplementary material. The engine and actuator response time and the minimum and maximum limits of the engine and actuator outputs are also provided in the supplementary material. This set of information must be appropriately considered in the physics engine.

The physics engine must be implemented in MATLAB/Simulink environment. The Graphics team will create the graphical environment in a way that it can communicate with MATLAB/Simulink the same way that FlightGear Flight Simulator does. Also, the User Interface team will create a joystick that will have a similar communication protocol to the ones support by MATLAB/Simulink joystick communication tools. Therefore, to verify the effectiveness of the Physics Engine you will need to connect your software to FlightGear and a joystick that will be provided and run free flight simulations.

You will need to divide the whole physics engine development into smaller parts, each to be handled by an individual. Set an internal deadline to complete the individual parts. Then, assign two other group members to review the individual's work to ensure correctness of all system modules.

Project Deliverables

This project is worth 35% of your final grade. The project is broken down into a number of project deliverables, detailed below with values and due dates.

1. Group Plan

Value: n/a, due date: start of lab 7

You are required to review the course material preferably before the lab and discuss with your group how to implement the equations in Simulink. You will need to create an objectives list/tree, a functional analysis (black-box, glass-box is recommended) and determine the means to implement each function. You should be able to sketch a block-diagram of the whole system. Assign each part to a student and select two other students to review the work. This is critically important to ensure every part is implemented correctly. Without the peer review, diagnosing the software will become challenging given the scale of the system. Set deadlines for completing individual tasks, completing the reviews, discussing possible flaws in each software part, and final integration of all parts. In setting the deadlines, keep in mind that the first implementation of software may have flaws, and you may need to have one or more iteration of developments. Therefore, allocate sufficient time to deal with potential pitfalls to deliver your software and report by the deadline. Create a summary of your plan, individuals assigned to each task and their reviewers, and your timeline, and submit the summary as your group plan before the start of lab 7. The group plan is mandatory to get a grade for the deliverables of the project.

2. Software Prototype

Value: 25%, due date: the end of lab 9

Every group is required to complete their physics engine in MATLAB/Simulink and connect it to a joystick and FlightGear software. In lab 9, you are expected to present a demo of flying a Boing 757 model over Downtown Toronto. All group members are required to attend the demo and they are expected to answer questions that the instructor and the TA may ask about how their software works.

3. Report

Value: 10%, due date: 48 hours after the end of lab 9

Along with the demo, you are required to submit a report about the details of your software design, and how to integrate it with a joystick and FlightGear. This report should include:

- The objective of the report, and an introduction of the design problem followed by the outline and organization of the report.
- A revised statement of the problem. It is expected that you will review the client statement and clarify it via consultation with your TA.
- Updated objectives list/tree.
- The name of contributors to each part of the design.
- A hierarchical block-diagram representation of the whole physics engine software starting with a high-level architecture and explanation of the data flow in the software.
- Description of all sub-systems in your software, explanations about the data flow in that sub-system, and details of any special settings for a Simulink block or a MATLAB command.
- A good practice is to write down the equations that each part of the software implements (if applicable).
- Presenting simulation results. Include plots for at least one simulation scenario where for a given initial condition, all the 12 states of the aircraft are predicted over a time window. The initial conditions will be provided.
- It is important that all the plots have a title and labeled axes. All plots must be clear. Using vector-based graphics (the method explained in the class using copy figure option of MATLAB) is mandatory.
- Conclusion: a summary of the work and implications, brief discussion of the advantages and disadvantages of the design and how it can be possibly improved in the future.

The report should be compiled into one document, logically organized, and clearly presented. Submissions will be through the course website.